

Metric-Driven Evaluation

Accuracy, Precision, and Recall in Data Science

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Confusion Matrix: Definition

A confusion matrix summarizes prediction results for binary classification:

$$\begin{bmatrix} \text{TP} & \text{FN} \\ \text{FP} & \text{TN} \end{bmatrix}$$

- TP: True Positives
- TN: True Negatives
- FP: False Positives
- FN: False Negatives

	model predic _(cancer)	Actual _(cancer)
True = 1		
False = 0		
TP	1	1
TN	0	0
FP	1	0
FN	0	1

The Foundation: Confusion Matrix

The confusion matrix categorizes predictions against actual ground truth labels.

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

- Type I Error (FP): The "False Alarm".
- Type II Error (FN): The "Miss".

Evaluating Global Performance: Accuracy

Accuracy measures the overall fraction of correct predictions.

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

When it fails: Consider a dataset with 990 Healthy people and 10 Sick people.

- A model that predicts "Healthy" for everyone gets **99% accuracy**.
- However, it fails to find any of the sick people (**0% Recall**).

Precision: Minimizing False Alarms

Precision answers: "When the model says POSITIVE, how often is it right?"

Formula

$$\text{Precision} = \frac{TP}{TP + FP}$$

Application Case: Spam Filtering

- High Precision is preferred here.
- A False Positive means a legitimate email is marked as spam (unacceptable).
- A False Negative means a spam email stays in the inbox (tolerable).

Recall: Maximizing Detection

Recall (Sensitivity) answers: "How many of the actual POSITIVE cases were found?"

Formula

$$\text{Recall} = \frac{TP}{TP + FN}$$

Application Case: Cancer Screening

- High Recall is mandatory.
- A False Negative means a sick patient goes untreated (disastrous).
- A False Positive means a healthy patient needs a second test (safe).

The Balancing Act: F1-Score

The F1-score is the **harmonic mean** of precision and recall. It penalizes extreme values of either metric.

Formula

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Why Harmonic Mean?

- Arithmetic mean of P=1.0 and R=0.0 is 0.5.
- Harmonic mean of P=1.0 and R=0.0 is **0.0**.
- It ensures the model performs reasonably well in *both* aspects.

Example Dataset

Email	Actual (Spam?)	Predicted (Spam?)
1	1	1
2	0	0
3	1	1
4	1	0
5	0	1
6	1	1
7	0	0
8	0	0
9	1	0
10	0	0

From this we get:

$$TP = 3, \quad FN = 2, \quad FP = 1, \quad TN = 4$$

Precision

Definition: Precision measures the proportion of predicted positives that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Precision} = \frac{3}{3 + 1} = \frac{3}{4} = 0.75$$

Interpretation: When the model predicts spam, it's correct 75% of the time.

Definition: Recall measures the proportion of actual positives that were correctly predicted.

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{3}{3 + 2} = \frac{3}{5} = 0.60$$

Interpretation: The model catches 60% of actual spam emails.

Definition: Harmonic mean of precision and recall.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F_1 = 2 \cdot \frac{0.75 \cdot 0.60}{0.75 + 0.60} = \frac{0.90}{1.35} \approx 0.667$$

Interpretation: Balanced measure that considers both false positives and false negatives.

Summary of Metrics

Metric	Value
True Positives (TP)	3
True Negatives (TN)	4
False Positives (FP)	1
False Negatives (FN)	2
Precision	0.75
Recall	0.60
F1 Score	0.667

Summary of Metrics

b. Consider the following confusion matrix:

TN = 400, TP = 200, FP = 150, FN = 40. Give your opinion on the results, focusing strongly on precision and recall. Justify your answer.

Precision

Formula:

$$Precision = \frac{TP}{TP+FP}$$

Calculation:

$$Precision = \frac{200}{200+150} = \frac{200}{350} = 0.571 \approx 57.1\%$$

Interpretation:

When the model predicts Positive, it is correct about 57.1% of the time.

This precision is not very high because the number of False Positives (150) is large.

Summary of Metrics

Recall

Formula:

$$Recall = \frac{TP}{TP+FN}$$

Calculation:

$$Recall = \frac{200}{200+40} = \frac{200}{240} = 0.833 \approx 83.3\%$$

Interpretation:

The model correctly detects about 83.3% of the actual positive cases.

This is a good recall because the number of False Negatives (40) is low.

Opinion on the Model

- The model has high recall but moderate precision.
- It can detect most positive cases successfully.
- However, it also produces many false positives.

Since FP = 150 is high, many negative cases are incorrectly classified as positive.